

KENYA PIPELINE COMPANY

REVENUE ASSURANCE — PROJECT MEMO

TO: KPC Revenue Assurance Stakeholders & PLP Supervisors
FROM: Null_Terminators Team
SUBJECT: Update — From KPC Revenue Leakage (Hackathon 1) to Inuka Foundation Governance (Hackathon 2)
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1. RECAP — HACKATHON 1: THE O2C REVENUE LEAKAGE PROBLEM

Kenya Pipeline Company (KPC) was losing revenue within its Order-to-Cash (O2C) cycle because operational data sat in silos, leaving it exposed to unmetered loading, invoicing/tariff errors, underpayments, and organised fraud such as product substitution and depot-shopping. FlowGuard's answer was to unify nine core datasets — physical (loading logs, inventory), commercial (dispatches), financial (invoices, payments), and master reference data (OMCs, products, depots, tariffs) — into a single three-way reconciliation engine: Dispatches → Invoices → Payments.

2. HACKATHON 2 — EXTENDING THE SAME ENGINE TO INUKA FOUNDATION

Rather than rebuild, Hackathon 2 upgraded the Stage 1 architecture into a Real-Time Unified Beneficiary Data Fabric. The same ETL pipeline, reconciliation logic, fraud graph, audit trail, and AI retrieval layer were re-pointed outward — from KPC fuel revenue to Inuka Foundation beneficiary stipends — proving the design is domain-agnostic, not a one-off build.

We selected Domain 5 — Data Governance, Security & Compliance — and solved both of its problem statements:

- Problem 9 — Beneficiary Data Privacy & Consent Management: a consent ledger written into the same hash-chained audit trail, automatic anonymisation for donor/public reporting, and full Kenya Data Protection Act (KDPA) alignment.
- Problem 10 — Automated Audit Trails & Financial Reconciliation: an immutable, hash-chained trail covering every beneficiary-record change, plus automated three-way matching of Attendance → Stipend Authorisation → Disbursement, with ghost payments, duplicates, and overpayments flagged critical by default.

3. THE LINKUP — ONE DATA FABRIC, TWO DIRECTIONS

A single flow_direction discriminator (inbound | outbound | all) lets one codebase, one schema, and one set of quality gates serve both missions:

Layer	Hackathon 1 — KPC (Inbound)	Hackathon 2 — Inuka (Outbound)
Reconciliation	Dispatches → Invoices → Payments	Attendance → Stipend Auth → Disbursement
Fraud graph	OMC ↔ Depot leakage networks	Officer ↔ Beneficiary + shared-account rings
Critical signals	Underpayment, overpayment	Ghost payments, duplicates, overpayments
Audit trail	Immutable hash-chain	Same chain, extended to consent & stipends
Roles	4 roles	5 roles (+ Inuka Manager, view-only)
AI assistant	Basic RAG on KPC docs	Bilingual (EN/SW) RAG on program & consent Q&A

4. QUANTIFIED IMPACT

Metric	KPC (Inbound)	Inuka (Outbound)
Reconciliation rate	95%+	95%+
Leakage / fraud detection	100% of anomalies detected	100% of ghost/duplicate/overpayment cases
Manual effort saved	80% (2 weeks/month → 2 days)	80% of support workload
Est. annual savings	~200M KES (4B KES revenue, 5% leakage)	~2.4M KES (staff hours saved)
Chatbot resolution	—	80% of tier-1 queries, <3 sec

Honesty of assumptions: savings are modelled estimates — 5% baseline leakage on 4B KES KPC throughput, and measured staff-hours reallocated during Inuka UAT — not guarantees.

5. QA, DEPLOYMENT & WHY THIS MATTERS FOR JUDGING

- 71% automated backend test coverage; 13 UAT scenarios, 100% pass rate; sub-200ms cached dashboard responses; CI/CD on every push.
- Manual verification of consent-withdrawal and anonymisation flows against KDPA requirements.
- Live: revenue-assurance.fly.dev (API/Swagger) and flowgardd.vercel.app (frontend).

The Linkup is direct evidence for "Technical Quality & Integration": the same ETL, reconciliation, fraud-detection, alerting, audit, and AI-retrieval framework was re-pointed at a new domain with no duplicated stack — turning a single hackathon build into a governed, fraud-resistant platform for both KPC revenue and Inuka Foundation disbursements.

CONCLUSION

Hackathon 1 proved the model on inbound KPC revenue. Hackathon 2 proves it generalises: the same data fabric now governs outbound Inuka Foundation stipends with full privacy, consent, and audit compliance. One architecture, two domains, one defensible source of truth against both revenue and disbursement leakage.